



17 June 2026, 15:45

Room Agora

Mapping Organizational Gradients of Cortical Growth

Using Spectral Normative Models

OHBM 2026 · Lifespan Development



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National University of Singapore
University of Melbourne



Disclosures

I have **no conflicts of interest** in relation to this presentation to disclose.



sina-mansour.github.io/OHBM_2026/oral

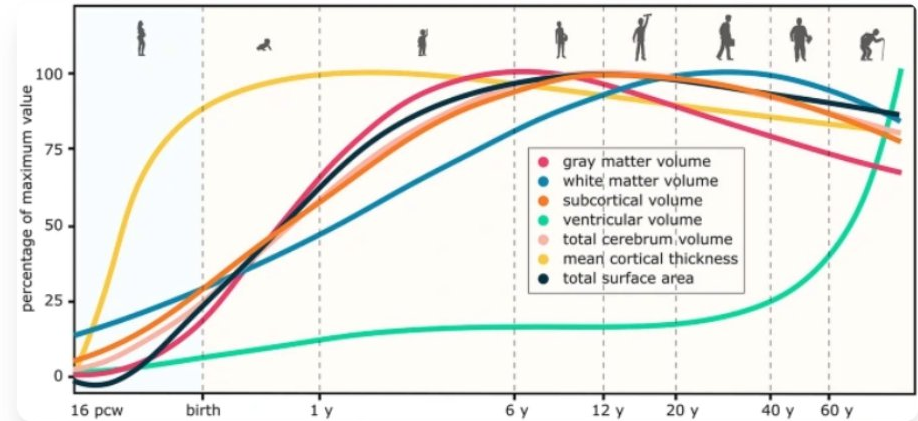
Cortical thickness changes across the lifespan

 Motivation

Cortical thickness changes across the lifespan

🧠 Motivation

- Thickness **peaks in early childhood**, then declines across the lifespan



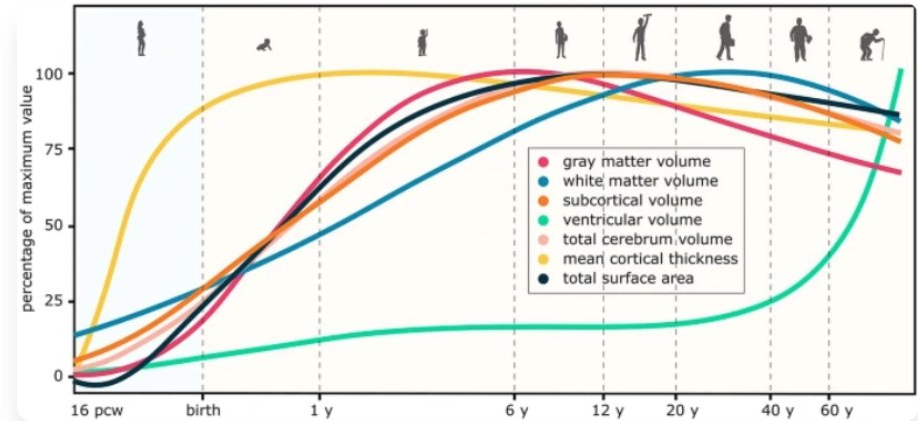
Bethlehem et al. (2022)

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Thickness **peaks in early childhood**, then declines across the lifespan

- Shaped by distinct processes: developmental **pruning** to aging **atrophy**



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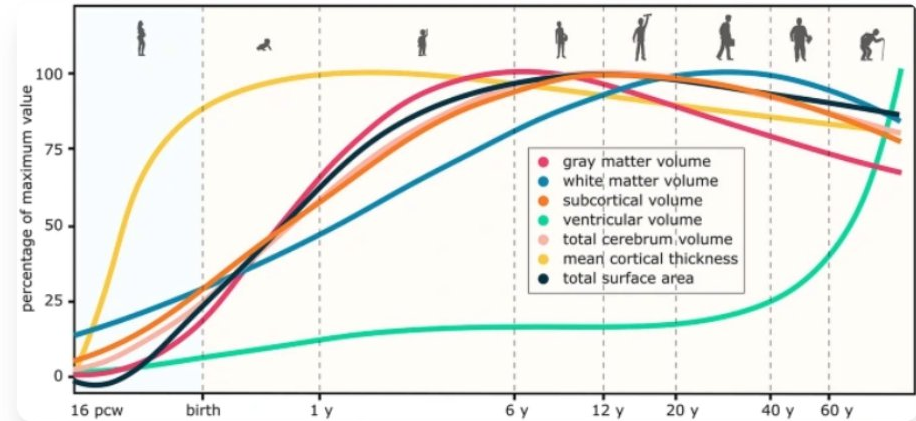
Cortical thickness changes across the lifespan

🧠 Motivation

Thickness **peaks in early childhood**, then declines across the lifespan

Shaped by distinct processes: developmental **pruning** to aging **atrophy**

➤ **Organised macroscopically across the cortex, and if so, *how?***



Bethlehem et al. (2022)

Cortical thickness mirrors cortical hierarchies

 Background & gap

Cortical thickness mirrors cortical hierarchies

 Background & gap

- The **group-average thickness map** mirrors established cortical organisational hierarchies

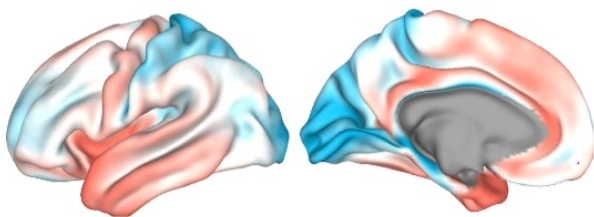
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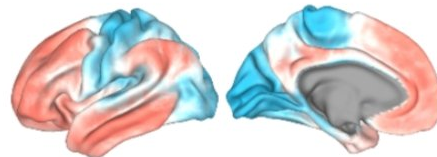
Cortical thickness

[Glasser et al. \(2016\)](#)



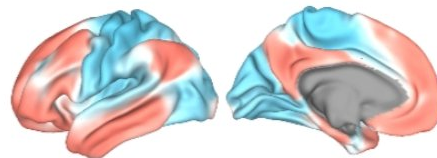
S-A axis

[Sydnor et al. \(2021\)](#)



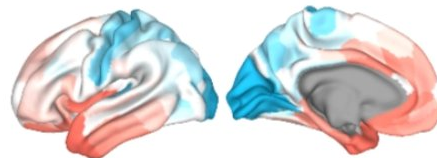
Functional gradient PC1

[Margulies et al. \(2016\)](#)



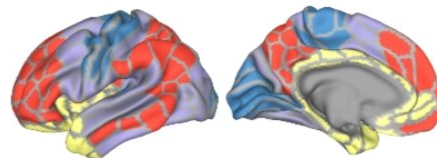
Gene expression PC1

[Burt et al. \(2018\)](#); [Dear et al. \(2024\)](#)



Cytoarchitecture

[Mesulam \(1998\)](#)



Cortical thickness mirrors cortical hierarchies

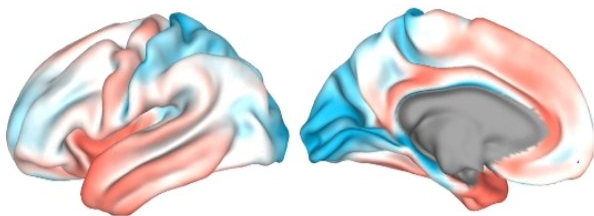
🔍 Background & gap

The **group-average thickness map** mirrors established cortical organisational hierarchies

➤ But how do these associations *change* across the lifespan?

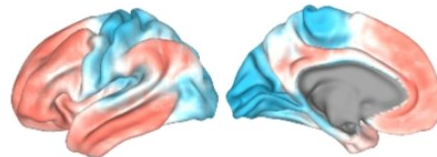
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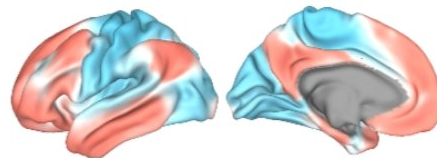
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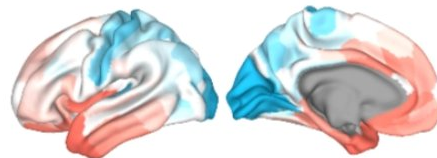
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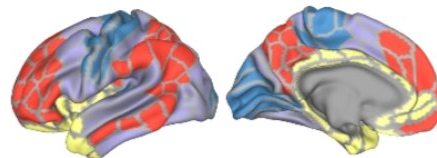
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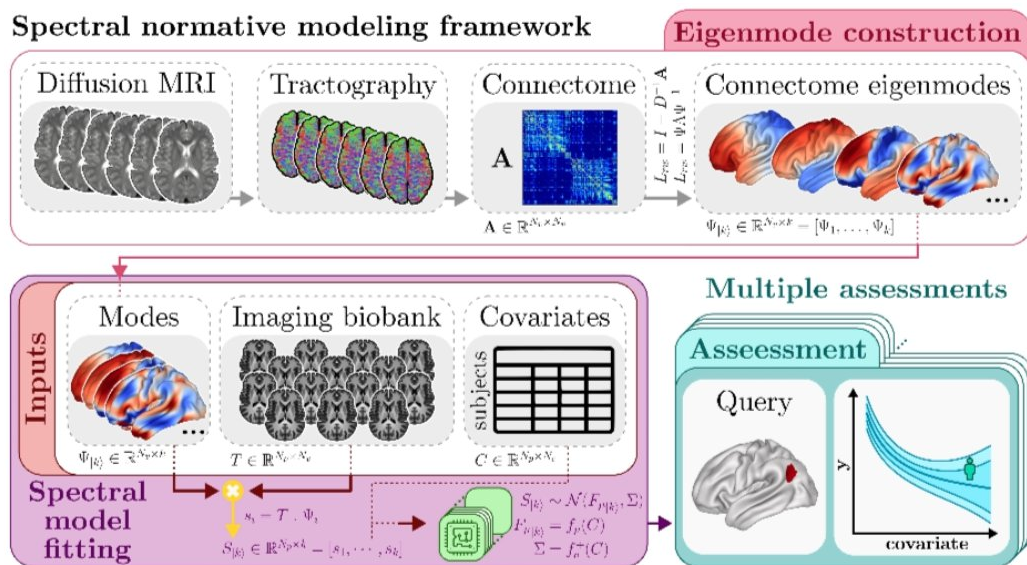
Spectral Normative Modelling (SNM)

⚙ Method

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⚙️ Method

- A **graph-spectral extension** of classical normative modelling, enabling efficient vertex-level inference of normative ranges



Spectral Normative Modelling (SNM)

⚙ Method

A **graph-spectral extension** of classical normative modelling, enabling efficient vertex-level inference of normative ranges

- Thickness encoded in a **connectome eigenmode** basis, then normative fitting that can be probed at arbitrary resolutions (any combination of vertices)

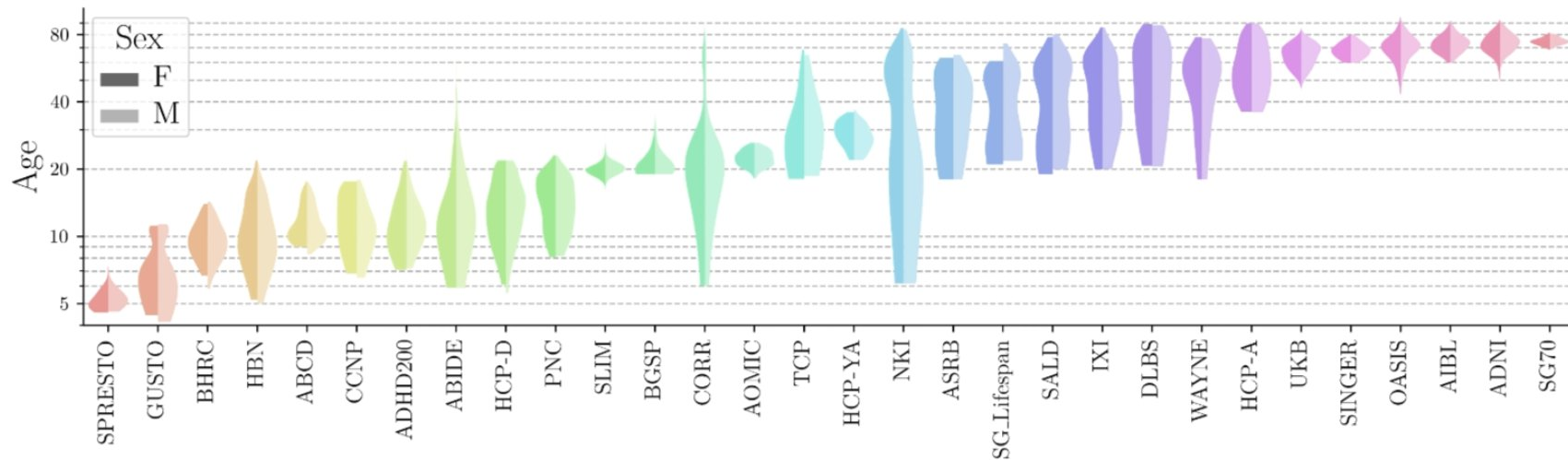
SNM training

 Structural MRI sample

SNM training

Structural MRI sample

➤ SNM was trained on a large, lifespan structural MRI sample

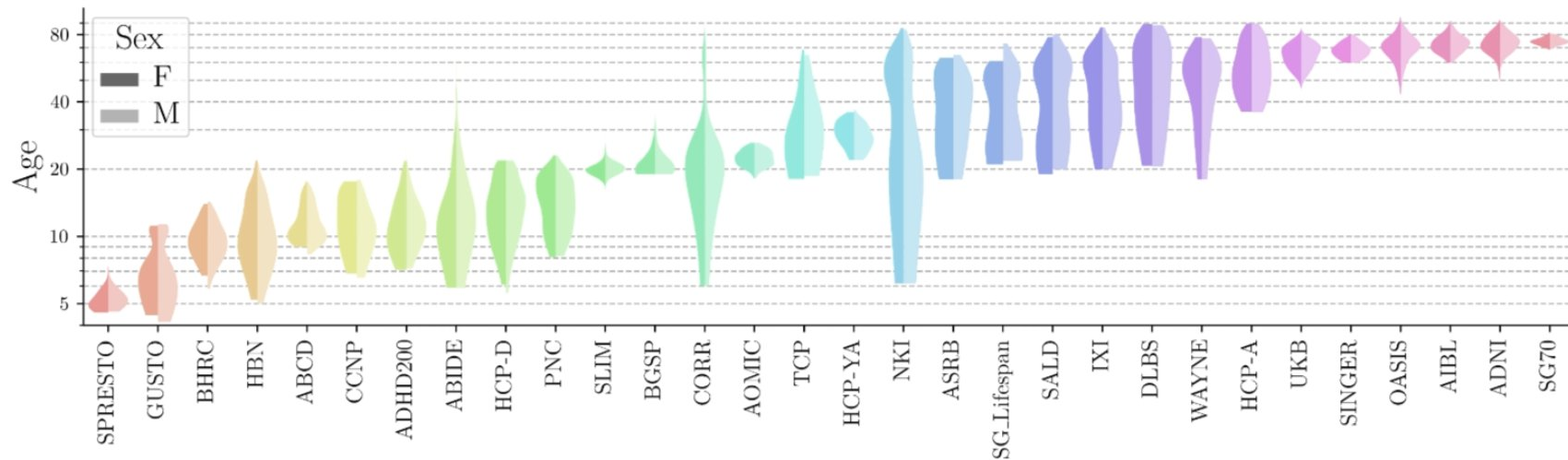


SNM training

Structural MRI sample

SNM was trained on a large, lifespan structural MRI sample

➤ **78,405 QC'd scans · 30 datasets · 189 sites · ages 5 to 95**



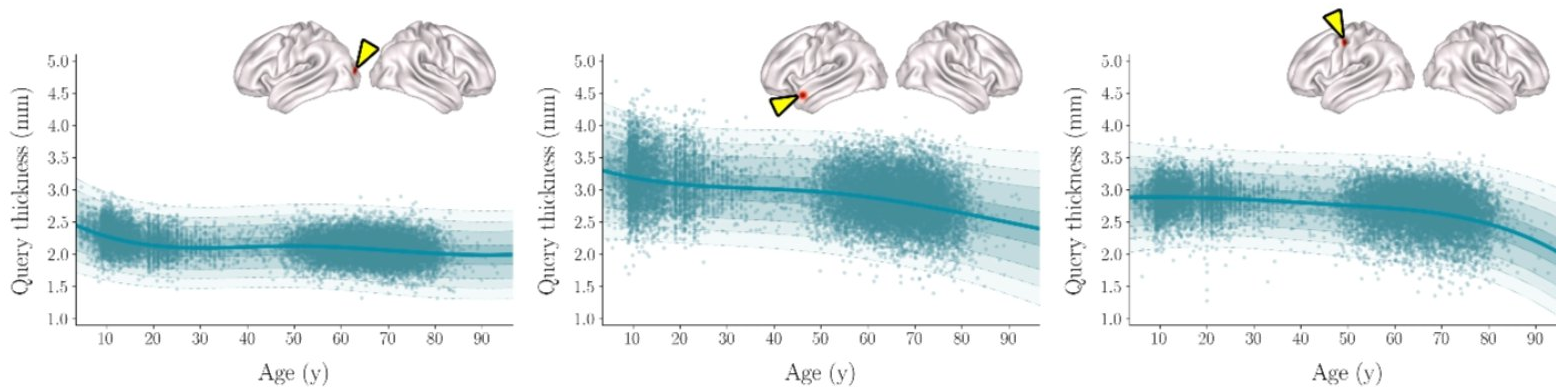
Vertex-resolution normative estimates

 SNM outputs

Vertex-resolution normative estimates

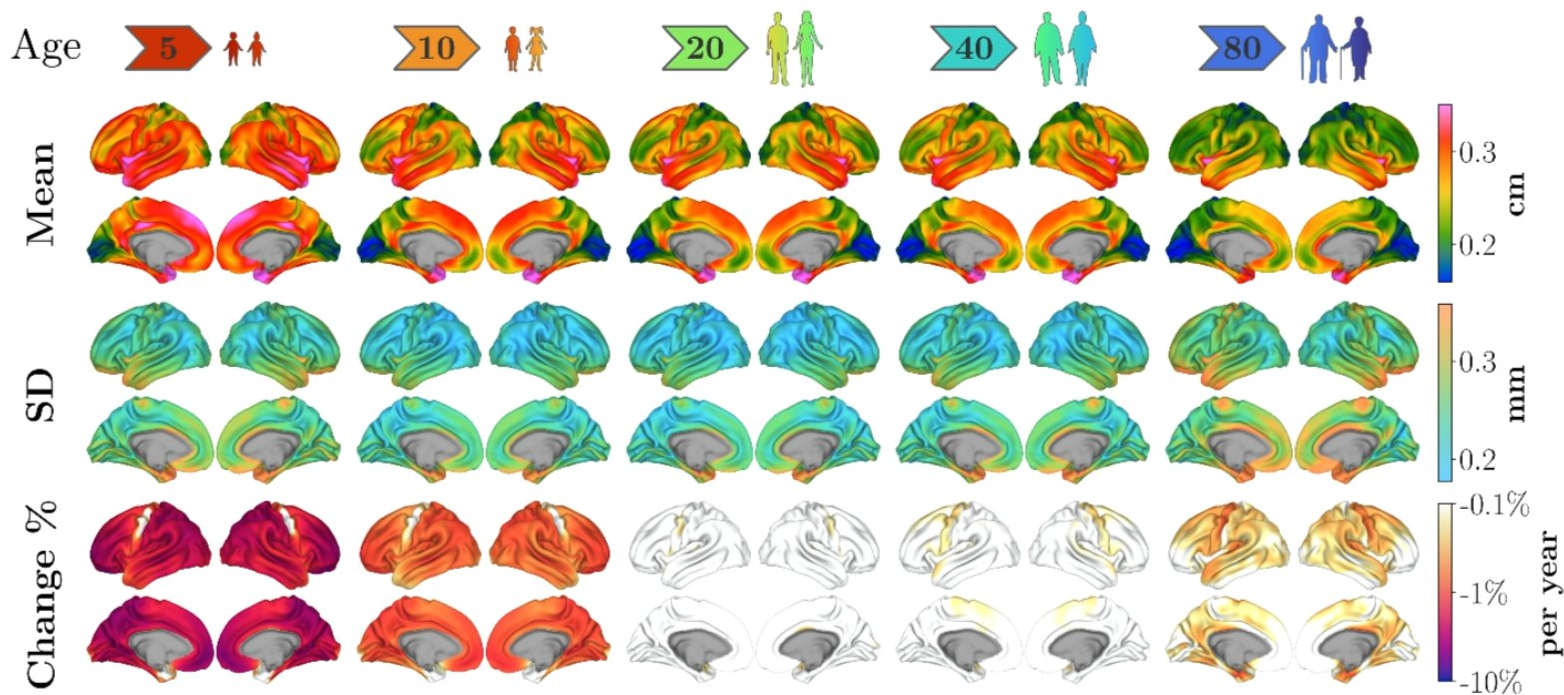
SNM outputs

- At any age, SNM estimates the **mean** thickness, **inter-individual variability**, and the **annual % rate of change** at every vertex.



Vertex-resolution normative estimates

SNM outputs



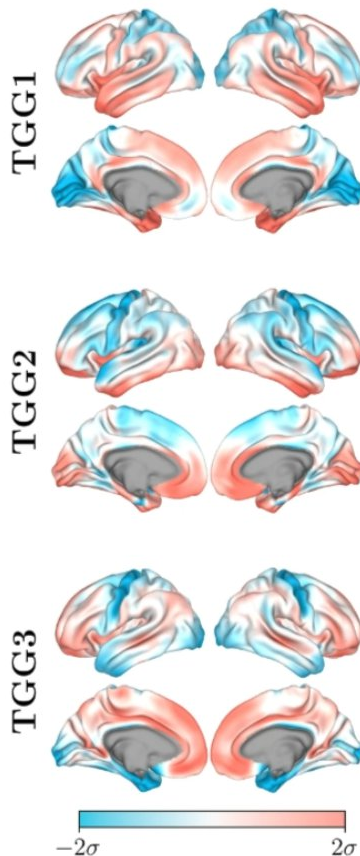
Thickness Growth Gradients

☀ From trajectories to gradients

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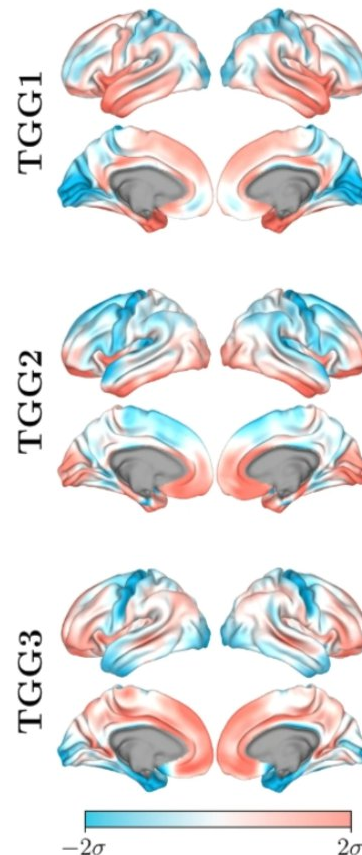
PCA on vertex trajectories: **3 gradients** explain **over 90%** of variance, the **Thickness Growth Gradients** (TGG1 to TGG3)



Thickness Growth Gradients

☀ From trajectories to gradients

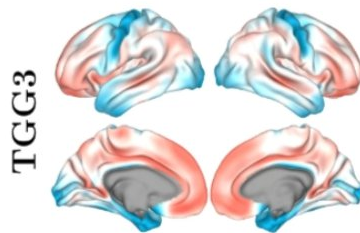
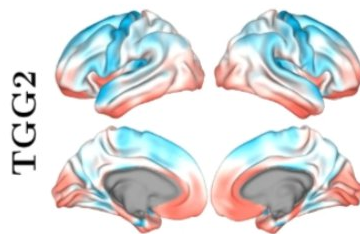
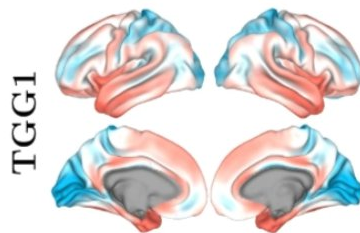
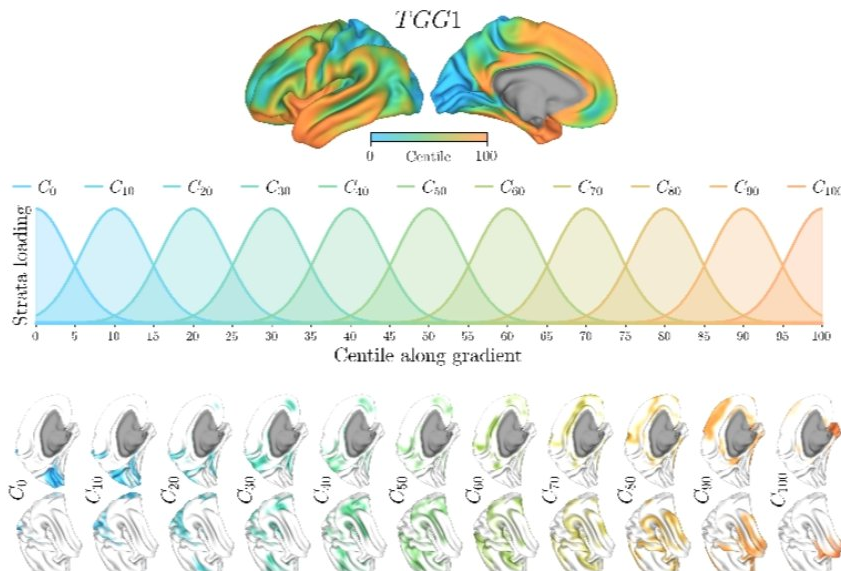
SNM answers **arbitrary post-hoc queries** on the trained model



Thickness Growth Gradients

☀ From trajectories to gradients

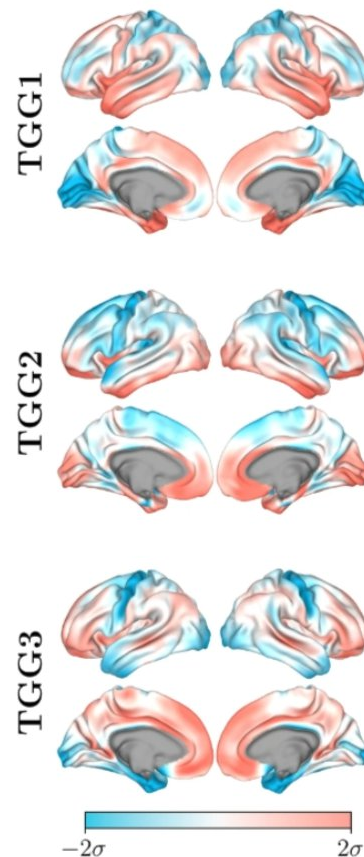
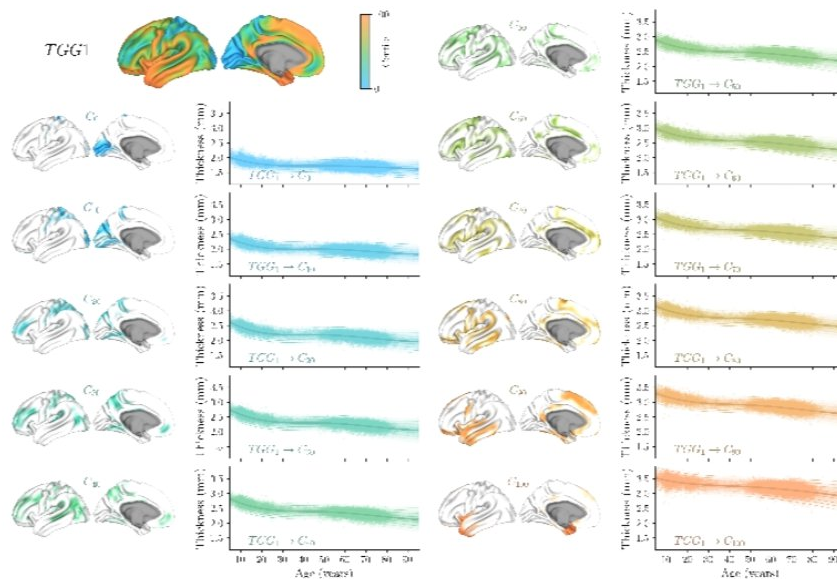
Define spatial **strata** along each gradient's centiles



Thickness Growth Gradients

☀ From trajectories to gradients

Query **normative trajectories** within each stratum



Thickness Growth Gradients

☀ From trajectories to gradients

Query **normative trajectories** within each stratum

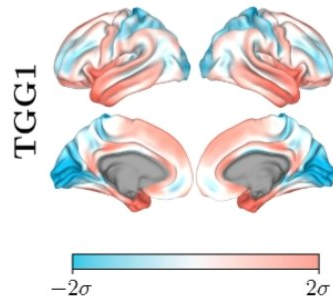
TGG1: stable global organisation

1 Age-invariant axis

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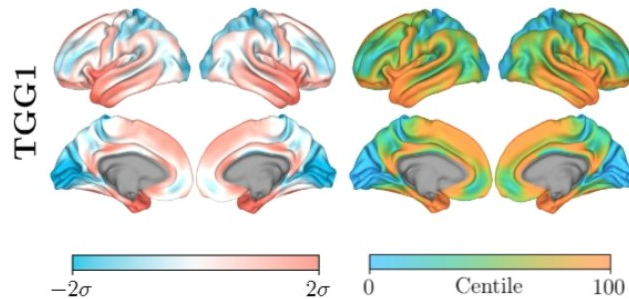
Values along **Thickness Growth Gradient 1**



TGG1: stable global organisation

1 Age-invariant axis

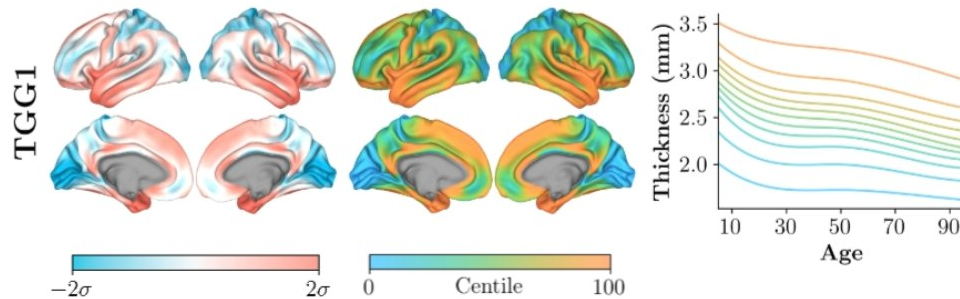
Converting values to **centiles**



TGG1: stable global organisation

1 Age-invariant axis

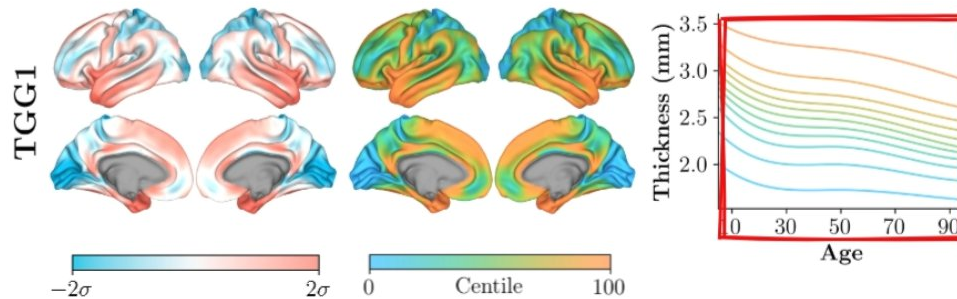
Mean thickness estimates for each stratum along the gradient



TGG1: stable global organisation

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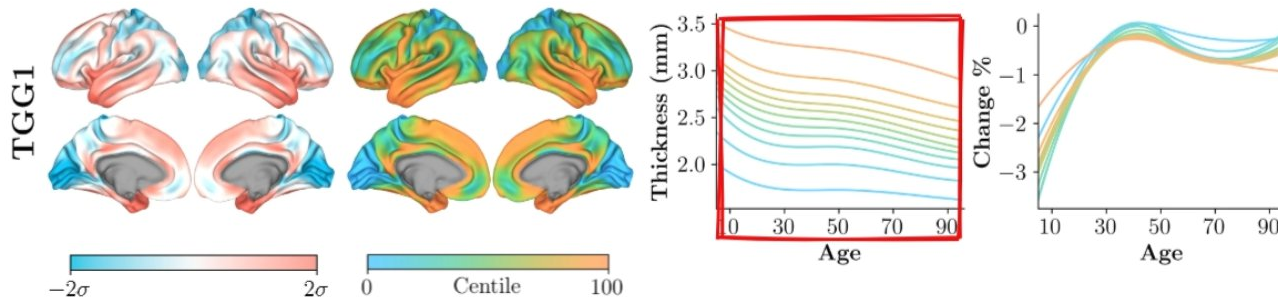
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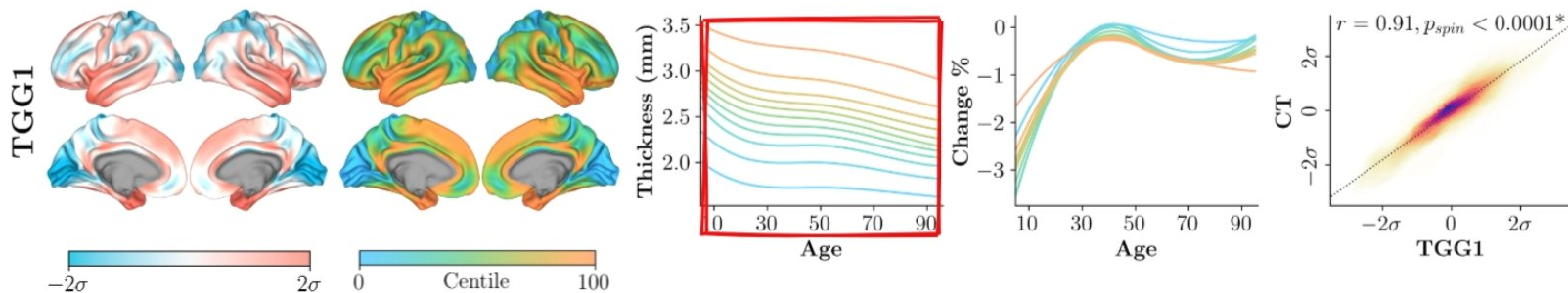
Percent change rate along the gradient strata



TGG1: stable global organisation

1 Age-invariant axis

Association with the **average cortical thickness map** ([Glasser et al. 2016](#))



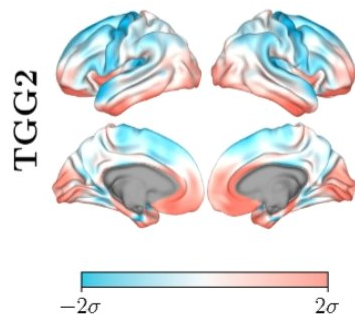
TGG2: aging-related thinning

2 Late-life decline

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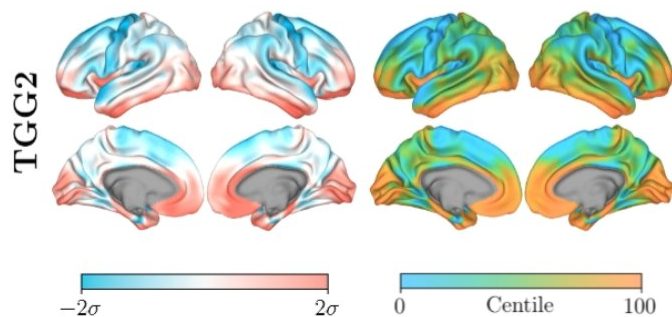
Values along **Thickness Growth Gradient 2**



TGG2: aging-related thinning

2 Late-life decline

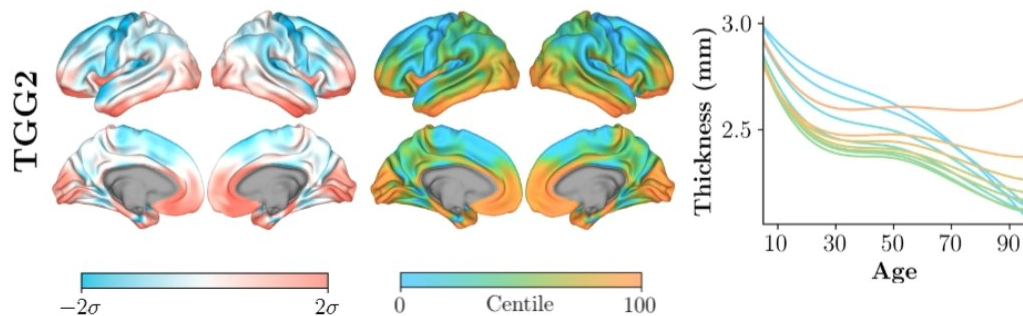
Converting values to **centiles**



TGG2: aging-related thinning

2 Late-life decline

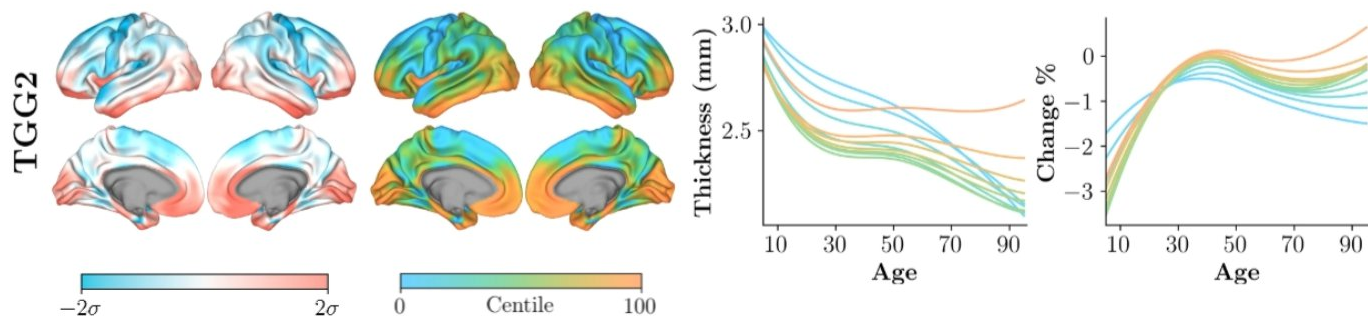
Mean thickness estimates for each stratum along the gradient



TGG2: aging-related thinning

2 Late-life decline

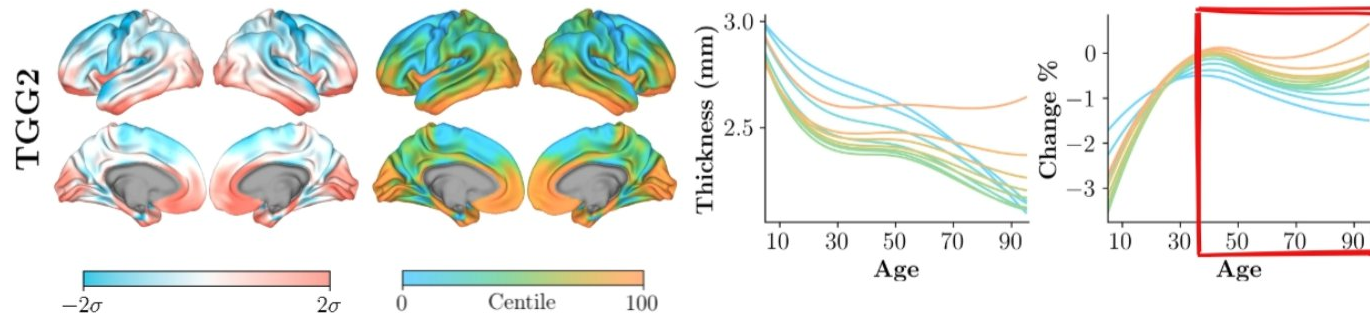
Percent change rate along the gradient strata



TGG2: aging-related thinning

2 Late-life decline

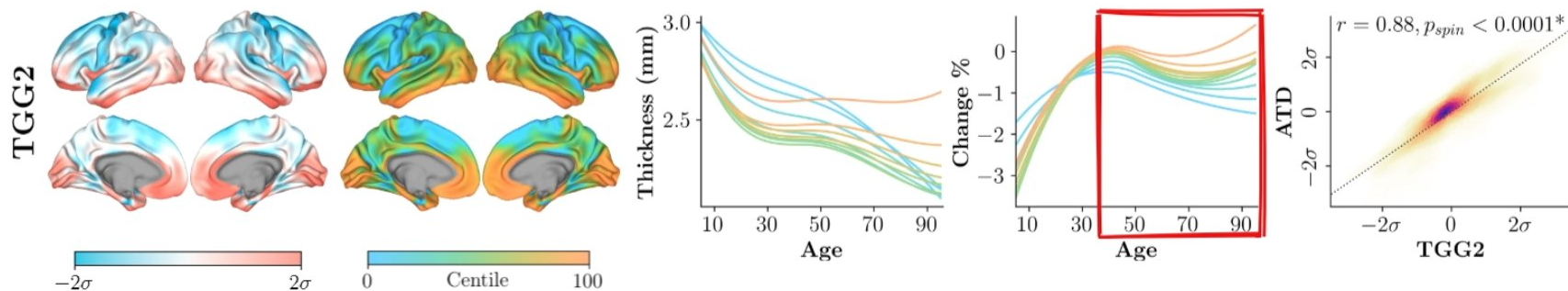
Percent change rate along the gradient strata



TGG2: aging-related thinning

2 Late-life decline

Association with **aging-related thickness decline**



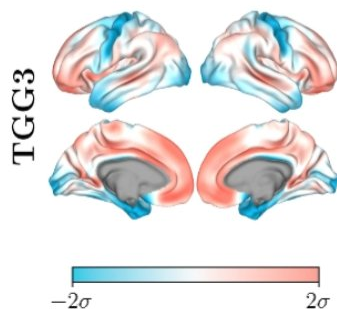
TGG3: developmental pruning

3 Early-life thinning

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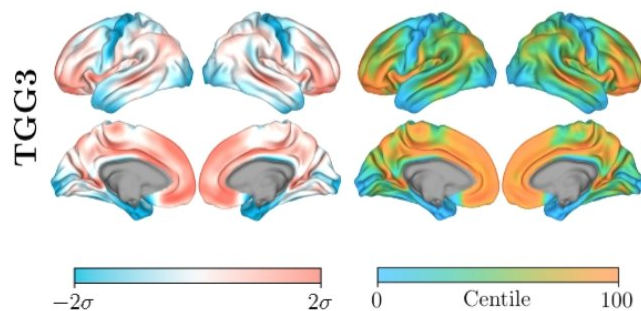
Values along **Thickness Growth Gradient 3**



TGG3: developmental pruning

3 Early-life thinning

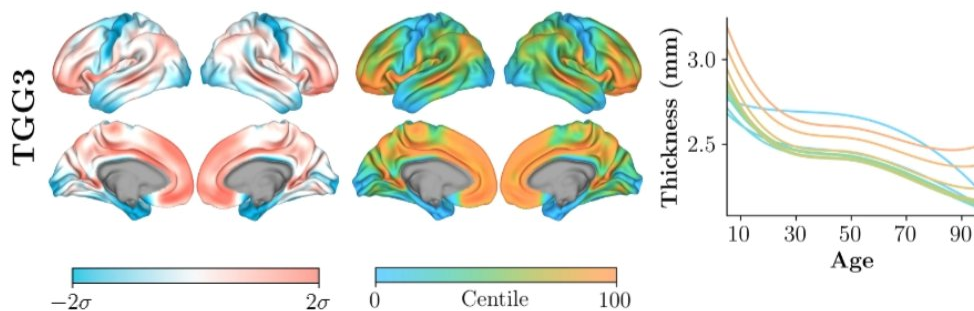
Converting values to **centiles**



TGG3: developmental pruning

3 Early-life thinning

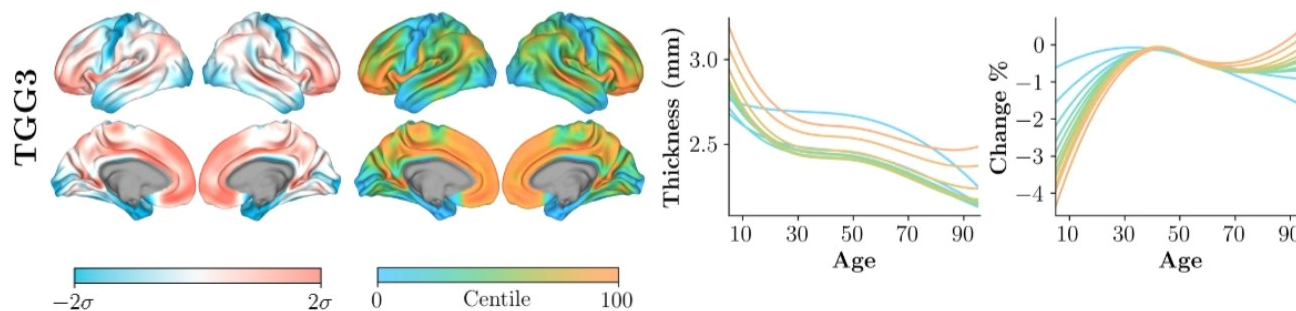
Mean thickness estimates for each stratum along the gradient



TGG3: developmental pruning

3 Early-life thinning

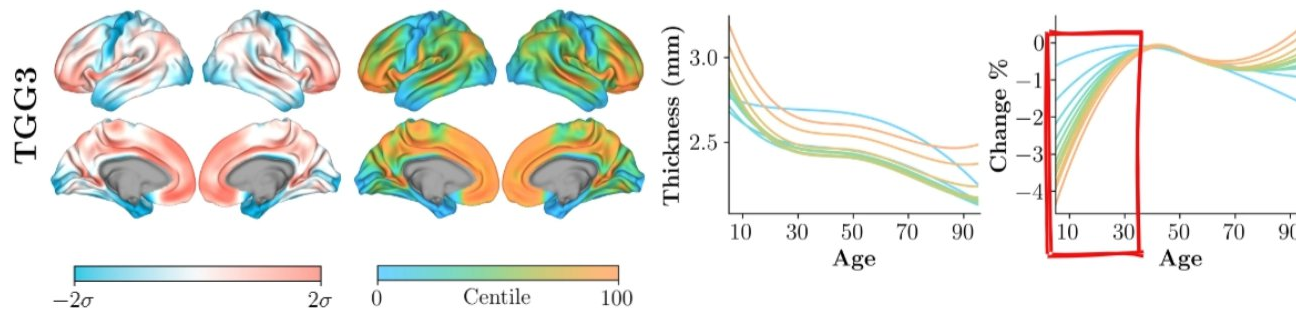
Percent change rate along the gradient strata



TGG3: developmental pruning

3 Early-life thinning

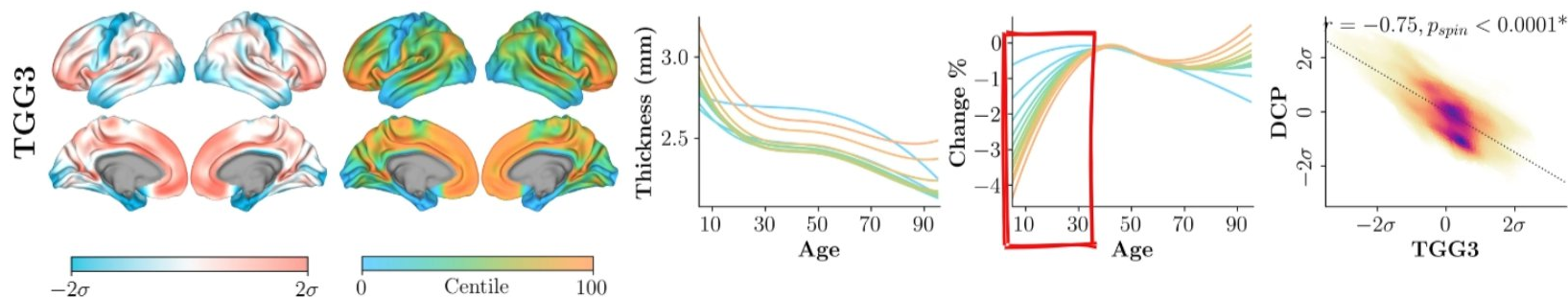
Percent change rate along the gradient strata



TGG3: developmental pruning

3 Early-life thinning

Association with **developmental cortical pruning**



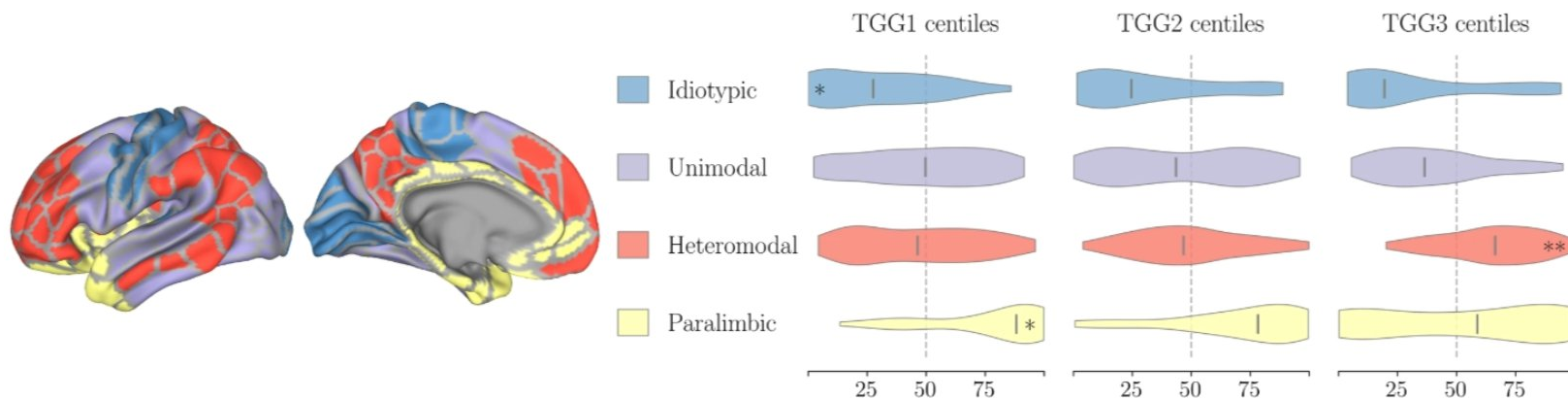
Cytoarchitectonic alignment

 From idiotypic to paralimbic

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 From idiotypic to paralimbic

➤ Each TGG aligns with the cortex's **laminar cytoarchitectonic hierarchy** ([Mesulam, 1998](#))



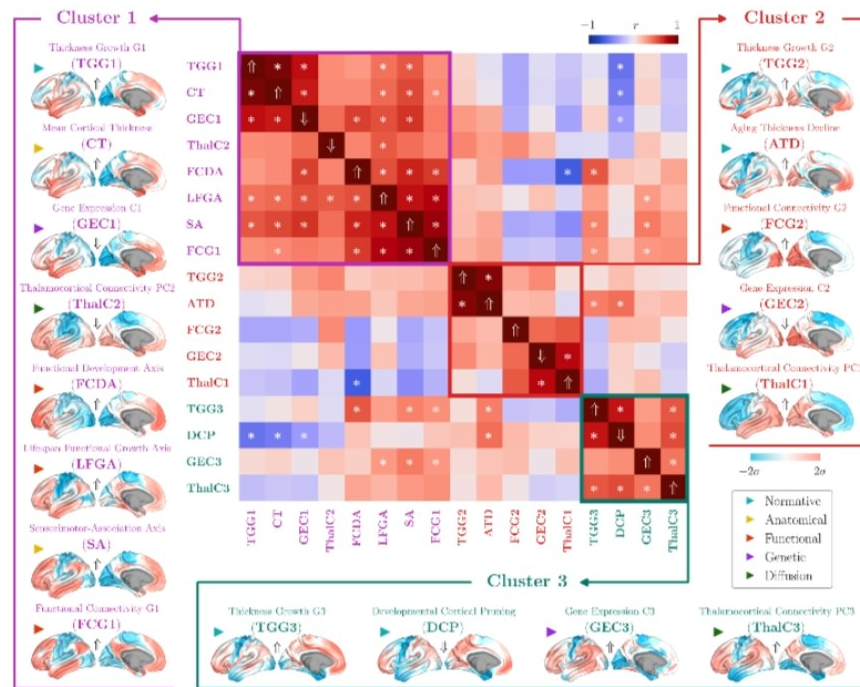
All three omnibus tests survive FDR correction using spatial-autocorrelation-preserving nulls.

Convergence with cortical hierarchies

✚ Colocating with established hierarchies across modalities

Convergence with cortical hierarchies

Colocating with established hierarchies across modalities



Normative retrogenesis

 Development vs. aging

Normative retrogenesis

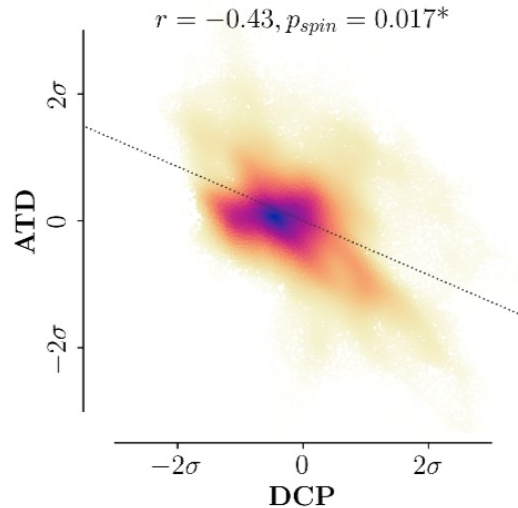
 Development vs. aging

- The cortical rates of normative thickness change **reverse patterns** in development vs. aging

Normative retrogenesis

↔ Development vs. aging

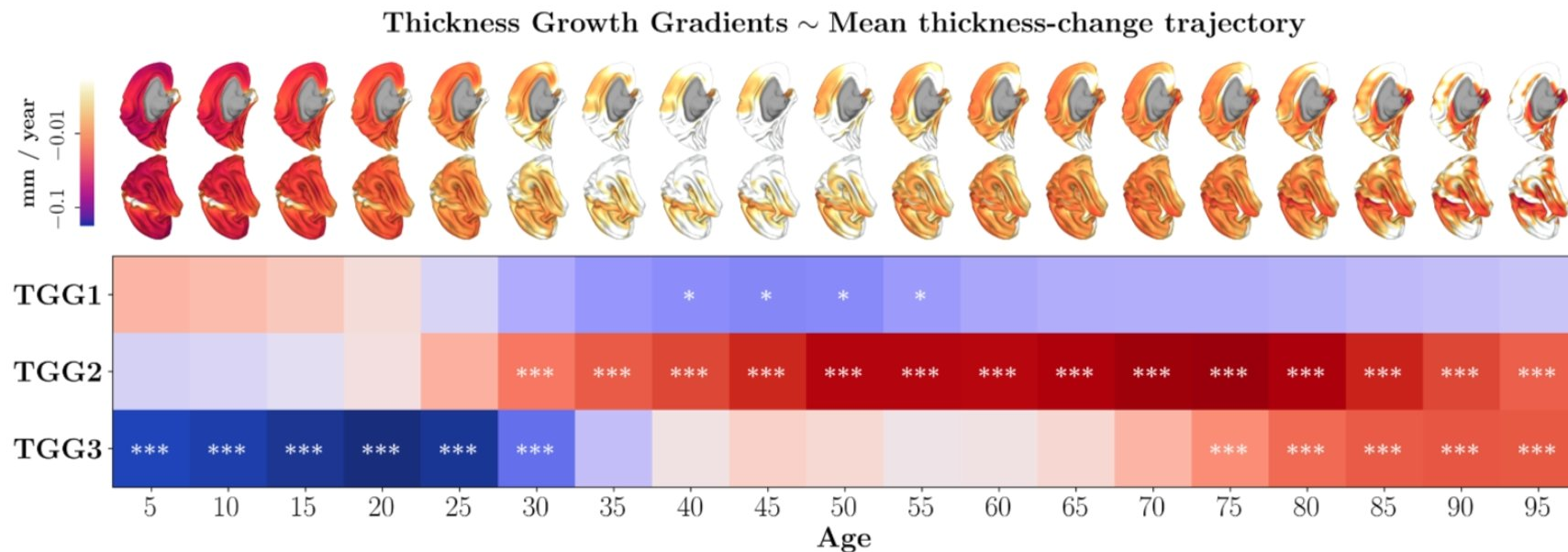
The cortical rates of normative thickness change **reverse patterns** in development vs. aging



Regions most **stable in development** are amongst those **steeply declining in aging**.

Normative retrogenesis

↕ Development vs. aging



TGG3 captures the aspects of development that reverse in aging.

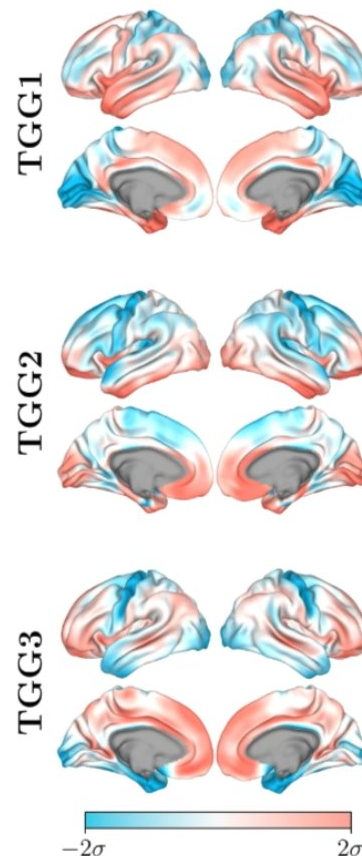
Take-home messages

 Conclusions

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- **SNM** makes lifespan vertex-resolution brain charting tractable

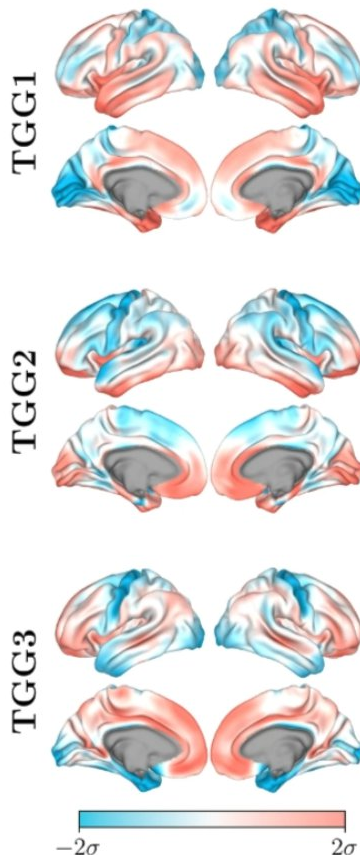


Take-home messages

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SNM makes lifespan vertex-resolution brain charting tractable

- Thickness change resolves into **three growth gradients** (TGG1 to TGG3)



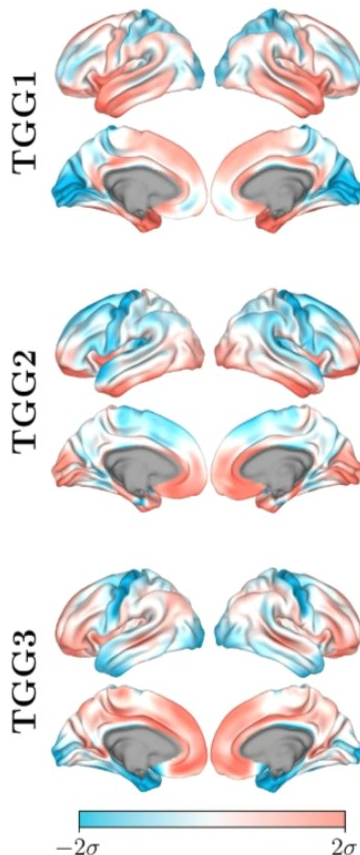
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- Each gradient recapitulates **established cortical hierarchies**: function, genetics, and cytoarchitecture



Take-home messages

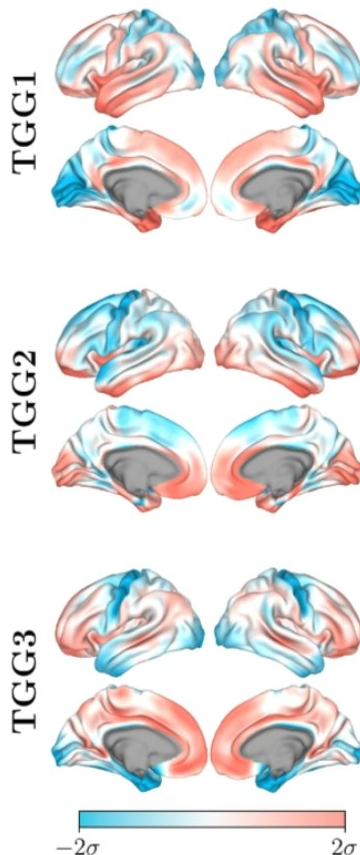
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Each gradient recapitulates **established cortical hierarchies**:
function, genetics, and cytoarchitecture

➤ **Development and aging are coupled by a normative *retrogenesis***



Code & Data Availability

⚡ Openly available resources

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Python Package

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⚡ Openly available resources



Python Package



GitHub Repository

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GitHub Repository



Preprint

Acknowledgements

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Florey

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sina-mansour.github.io/OHBM_2026/oral